

BIG DATA PROCESSING.

Lab01: Python Refresh

BACKGROUND.

The file **exercises.py** contains the 7 functions to be completed.

- **ex1:**
The function prints your name by the screen.
Example: In my case it will print Nacho Castineiras
- **ex2:**
The function declares a new variable res, assigns it to the sum of 'a' and 'b' and returns res.
Example: If a = 3 and b = 5 then it returns 8 (which is 3 + 5)
@param a: First Integer
@param b: Second Integer
@return Sum of a and b
- **ex3:**
The function receives 3 numbers and prints by screen the biggest of them.
Example: If a = 3, b = 7 and c = 5, then it prints 7 (which is the biggest of the 3 numbers).
@param a: First number
@param b: Second number
@param c: Third number
- **ex4:**
The function returns the sum of all numbers from 1 to n.
Example: If n = 5, the function returns 15 (which is 1 + 2 + 3 + 4 + 5).
@param n: Number we want to stop adding at
@return Sum of all integers in [1..n]
- **ex5:**
The function prints a pattern by screen.
Example1: If n = 3, then it prints
*
**

Example2: If n = 5, then it prints<

```
*  
**  
***  
****  
*****
```

@param n: Number of lines to be printed

- **ex6:**

The function reverses a String and returns the String reversed.

Example: If the String "Hello" is received, then it returns "olleH"

@param s: String to be scanned.

@return The reversed String.

- **ex7:**

NOTE: This exercise has been taken from CodeWars:

<https://www.codewars.com/kata/sum-of-digits-slash-digital-root>

Description:

A digital root is the recursive sum of all the digits in a number.

Given n, take the sum of the digits of n.

If that value has still more than one digit, continue reducing in this way until a single-digit number is produced.

Example 1:

ex7(16)

1 + 6

7

Example 2:

ex7(942)

9 + 4 + 2

15

However, as 15 still contains more than one digit, we iterate again

1 + 5

6

@param n: Number to apply its digital root to.

@return res: Digital result of the number.

EXERCISE.

Implement the 7 functions of exercises.py